

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

**THIS MONTH IN SCIENCE:** REDUCING COLON CANCER MORTALITY IN INDIA, THE FIRST INTERSTELLAR COMET AND THE BIRTH OF THE HIMALAYAS.



## Exoplanets as instruments

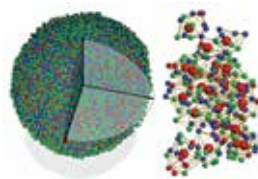
Precise measurement of the temperature of the surface of exoplanets could be used to infer the effect because of dark matter, and indicate the presence of dark matter, essentially making exoplanets dark matter detectors.

<http://dgit.in/xodmdet>

### WHAT'S NEW

## Atomic structure of amorphous solid probed for first time

**R**ecap of some elementary school stuff, there are two kinds of solids, crystalline and amorphous. The regular repeated patterns in which atoms arrange themselves in crystalline solids such as silicon or quartz are well understood and well studied. Scientists had little understanding of the three dimensional structures within amorphous solids such as plastics, glass, gels and rubber. Till now. The team used a technique they developed known as atomic electron tomography where electrons are passed through a sample to generate an atomic image



of a slice of the sample. Then the sample is rotated, and the process is repeated. After 55 rotations, all the

slices are put together, using image analysis and algorithms to form a 3D image of all the 18,000 atoms in the sample of metallic glass 8 nanometers in diameter. The study indicates that some atoms form regular clusters, showing that the distribution is not totally random.

<http://dgit.in/amosol>

## Wireless powering of IoT devices using 5G networks

**R**esearchers from the Georgia Institute of Technology have come up with an innovative way to harness this extra capacity, in the form of a rectenna that is capable of harvesting energy at 28-GHz band. The researchers used a Rotman lens, which is normally used in RADAR instruments for detection without having to turn the antenna, to make the rectifying antenna generate electricity have a wider angle



of coverage. The problem overcome by the researchers was that rectennas have wide apertures and a narrow field of view, which meant that they had to be close to, and aimed directly at the 5G base

station. By incorporating the Rotman lens for beamforming, the rectenna could generate electricity even at distances and when not aimed directly at the base station. This allows a device powered by the rectenna become direction agnostic for charging up. So far, the FCC limits on power and radiation prevented the use of such technologies, but it will be possible on 5G networks.

<http://dgit.in/5gpow>



### Don't get news on FB

In a study, those who trusted and consumed information about the pandemic from news channels and social networks were shown to give more incorrect answers to questions related to the pandemic, as against those who got their news from official government websites.

<http://dgit.in/msa>



### Dogs can smell the virus

Dogs are apparently incredibly effective at sniffing out the virus say scientists who trained the dogs specifically for that task. The dogs could identify the positive samples with a staggering 96 percent accuracy. The team is working on a tshirt smell test.

<http://dgit.in/doggottest>



### Ninja variant in 15 countries

An undetected variant of the novel coronavirus had spread to fifteen countries before it was spotted. The variant B117 (501Y) seems to have originated in the US in October, spread to UK and Europe by November and had even reached India before being discovered.

<http://dgit.in/covb117>